

PEOPLE ANALYTICS

Hire-to-Retire Analytics Platform

From First Application to Final Day — A Complete, Data-Driven View of the Employee Journey

Project Type	End-to-End People Analytics Solution
Tech Stack	Python · SQL Data Warehouse · Power BI
Dataset Scope	750 Employees 6-Year Lifecycle (2019–2024)
Document Version	1.0 Portfolio Edition

1. Executive Summary

"Every organisation is, at its core, a story of people — how they are found, how they grow, and why they stay. This project turns that story into evidence."

The Hire-to-Retire Analytics Platform is an end-to-end People Analytics solution that follows an employee's entire journey — from the moment a job requisition opens, through hiring, day-to-day performance, engagement, and eventually departure or retirement. It was built to answer a simple but powerful business question: what truly drives a productive, engaged, and loyal workforce — and where is the organisation quietly losing value?

Rather than treating HR data as a static spreadsheet, this project transforms it into a living decision-support system. Raw, messy source data is cleaned and structured through a Python-based pipeline, organised into a well-governed data warehouse, and finally brought to life through an interactive Power BI experience spanning five focused dashboards. The result is a single source of truth that lets leadership move from "what happened" to "what should we do next."

At its heart, the platform is built around a Quality-Cost-Speed philosophy: it recognises that hiring quickly, hiring cheaply, and hiring well are three different goals that often pull in different directions — and it gives decision-makers the visibility to balance all three deliberately, rather than by accident.

Why This Project Matters

- Replaces gut-feeling HR decisions with a structured, evidence-based view of the workforce.
- Connects recruitment choices to long-term business outcomes such as performance, engagement, and retention.
- Surfaces hidden problems — such as inconsistent management quality or below-market pay — before they turn into costly resignations.
- Demonstrates a complete, professional analytics workflow: from raw data to cleaned data, to a governed warehouse, to an executive-ready dashboard.

2. Dataset Overview

The platform is powered by a rich, six-year synthetic People Analytics dataset representing 750 employees, designed to mirror the complexity of a real mid-size organisation. It captures four connected chapters of the employee story, linked together through shared employee and candidate identifiers so that a single person can be traced from their very first job application through to their final day.

750 Employees Tracked	2019–2024 Years of History	~7,500 Performance Records	4 Connected Data Sources
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The Four Pillars of the Employee Story

- **Recruitment** — how each employee was sourced, how long it took to hire them, what it cost, and their quality profile at the point of hire (skills, cultural fit, and future potential).
- **Employee Master** — the core personnel record: role, department, salary, and the manager each employee reports to.
- **Performance Timeseries** — a quarterly scorecard tracking how each employee's output, work quality, and skill growth evolve over time.
- **Lifecycle & Retention** — the engagement and exit story — satisfaction trends, promotions, pay competitiveness, and whether and why an employee eventually left.

A Note on Data Quality

The dataset intentionally reflects real-world imperfections rather than an idealised picture. It builds in realistic effects such as manager influence on performance scores and the protective role that strong cultural fit can play for otherwise lower-skilled hires. These effects make the analysis meaningfully close to what a practitioner would encounter with genuine company data, and they are precisely what the dashboards are designed to bring to the surface.

Important: this is a synthetic, illustrative dataset created for demonstration and learning purposes. All names, identifiers, and figures are simulated and do not represent real individuals or a real company.

3. System Architecture

The platform follows a clean, linear flow that takes data from its raw, unstructured state all the way to an interactive business dashboard. Each stage has a single clear responsibility, which keeps the system easy to trust, maintain, and extend.

Stage 1 — Raw Data Intake

The journey begins with a single source workbook containing five raw data sheets (Recruitment, Employee Master, Performance, Lifecycle & Retention, and Monthly Engagement Pulse). This is the organisation's unfiltered, real-world data — complete with the missing values, inconsistent formats, and duplicate entries typical of any live HR system.

Stage 2 — Python Data Cleaning

A series of dedicated Python notebooks, one per data source, systematically prepares the data for analysis. For each source, the process:

- Profiles the data — checking its structure, size, and data types.
- Identifies gaps and inconsistencies — such as missing hire dates, blank exit reasons, or absent performance scores.
- Applies sensible business logic to fill gaps — for example, marking employees without an exit date as "Still Active", or filling missing performance scores with a representative average.
- Standardises data types and removes duplicate records to ensure consistency.
- Exports a clean, analysis-ready version of each dataset for the next stage.

Stage 3 — The Data Warehouse (Medallion Architecture)

The cleaned data is then loaded into a structured SQL data warehouse, organised using an industry-standard four-layer "medallion" approach. Each layer refines the data a little further, creating a transparent and auditable trail from raw input to business-ready output.

Layer	Purpose
Staging	Receives the cleaned files exactly as they arrive, with flexible, text-based columns to avoid load failures.
Bronze	Creates a timestamped, permanent record of the data exactly as it was received — the organisation's audit trail.
Silver	Applies proper data types, business keys, and validation rules, turning raw records into trustworthy, structured tables.
Gold	Reshapes the data into a business-friendly model — a set of clear "employee facts" and "reference lists" (department, manager, source, exit reason, dates) — purpose-built for fast, reliable reporting.

Stage 4 — Power BI Reporting Layer

The Gold layer feeds directly into Power BI, where the data is transformed into a five-page interactive dashboard experience. Business-friendly measures — such as average time to hire, attrition rate, and average engagement score — are defined once and reused consistently across every page, ensuring every number on every screen tells the same, trustworthy story.

End-to-end flow: Raw Workbook → Python Cleaning → Staging → Bronze → Silver → Gold → Power BI Dashboards

4. Key Insights & Dashboard Highlights

The Power BI experience is organised into five purpose-built pages, each answering a distinct business question. Every page opens with a set of at-a-glance KPI cards, followed by visuals designed to explain the "why" behind the numbers.

4.1 Overview — The Executive Snapshot

A single-page summary designed for leadership, giving an instant pulse-check on the health of the entire workforce.

Avg. Days to Fill Hiring Speed	Avg. KPI Achievement Workforce Performance	Avg. Pulse Score Employee Engagement	Attrition Rate Retention Health
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- Organisational Engagement Trend — tracks how company-wide engagement has moved over time, offering an early warning system for culture shifts.
- Departmental KPI Performance — compares performance across departments, quickly surfacing high performers and areas needing attention.
- Employee Age Distribution — gives a snapshot of workforce demographics for succession and workforce-planning conversations.
- Attrition Risk Profile by Department — flags which departments carry the greatest flight risk today.

Business takeaway: leadership can answer "how are we doing right now?" in under thirty seconds, without opening a single spreadsheet.

4.2 Recruitment — The Cost and Speed of Building a Team

This page evaluates how effectively the organisation finds and onboards talent, and at what cost.

Avg. Days to Fill Time to Hire	Avg. Acquisition Cost Cost per Hire	Hire Rate Pipeline Conversion	Avg. Quality Hired Talent Quality
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- Hiring Velocity (Days to Fill Trend) — shows whether the organisation's hiring speed is improving or slowing over time.
- Acquisition Cost by Sourcing Channel — compares channels such as referrals, agencies, and job boards to reveal which deliver the best value.
- Recruitment Pipeline Funnel — visualises how candidates move from application to hire, highlighting where the pipeline leaks.

Business takeaway: the data consistently shows that the fastest hiring channel is rarely the most cost-effective one — quality-first channels such as referrals tend to deliver stronger long-term value even when they aren't the quickest.

4.3 Performance — Are Employees Growing and Delivering?

This page connects the quality of a hire to what they actually deliver once on the job.

Avg. KPI Achievement Target Delivery	Avg. Quality of Work Output Quality	Avg. Skill Acquisition Learning Speed	Top Department Best Performer
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- Monthly KPI Achievement Trend — tracks whether performance is improving quarter over quarter across the organisation.
- Quality vs. Skill Acquisition Matrix — plots employees by underlying quality against how quickly they are developing new skills, helping identify future high-potential talent.
- Detailed performance breakdown table — allows a granular, department-by-department drill-down for people managers.

Business takeaway: performance scores should never be read in isolation — the platform is explicitly designed to help distinguish genuine underperformance from the effect of inconsistent management quality.

4.4 Lifecycle & Retention — Why People Stay, and Why They Leave

This page shifts the focus from performance to loyalty, identifying the earliest signals of flight risk.

Attrition Rate Workforce Turnover	Avg. Tenure (Months) Employee Longevity	Avg. Attrition Risk Predictive Flight Risk	Avg. Comp vs Market Pay Competitiveness
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- Compensation vs. Attrition Risk Correlation — visually tests whether employees paid below market rate are genuinely more likely to leave.
- Departmental Flight Risk Analysis — ranks departments by their exposure to turnover, guiding where retention efforts should focus first.
- Primary Drivers of Employee Turnover — a treemap breaking down exit reasons, separating voluntary departures from involuntary and retirement exits.

Business takeaway: compensation and cultural fit consistently emerge as the strongest early warning signs of attrition — well ahead of performance scores alone.

4.5 Pulse & Engagement — The Emotional Health of the Company

The final page tracks the ongoing sentiment and satisfaction of the workforce, the leading indicator behind almost every other metric on the previous pages.

Avg. Pulse Score Overall Engagement	High Culture Band % Strong Cultural Fit	Lowest Pulse Month Risk Period	Avg. Pulse by Dept. Team-Level View
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- Company Engagement Trend — the long-term pulse of employee sentiment across the entire observation period.

- Workforce Distribution by Culture Band — segments employees into engagement tiers, from thriving to at-risk.
- Engagement Sentiment by Department — pinpoints exactly which teams need attention first.

Business takeaway: engagement is a leading indicator, not a lagging one — teams with declining pulse scores today are tomorrow's attrition and performance risk.

5. Conclusion & Recommendations

The Hire-to-Retire Analytics Platform demonstrates that the full employee journey — sourcing, hiring, performing, engaging, and eventually leaving — can be captured, connected, and understood through a single, coherent analytics system. By pairing disciplined data engineering (Python cleaning and a governed SQL warehouse) with an intuitive Power BI experience, the project turns fragmented HR records into a genuine decision-making asset.

Most importantly, the platform is built to challenge easy assumptions. It doesn't simply report that performance is low or that someone has left — it equips the analyst to ask why, and to separate genuine talent issues from organisational ones such as management quality or below-market pay.

Recommendations for Ongoing Use

- **Review flight-risk signals monthly**, not annually — pulse score and compensation-versus-market are the earliest and most reliable warning signs available in this dataset.
- **Treat low performance as a question, not a verdict** — always check whether a manager effect could be influencing the numbers before making people decisions.
- **Favour quality-adjusted hiring channels**, even when they are not the fastest, since they tend to deliver stronger long-term value.
- **Protect cultural fit during recruitment**, as it consistently proves to be one of the strongest predictors of long-term retention.
- **Extend the model over time** by feeding in new quarterly and monthly data through the same pipeline, keeping the dashboard continuously current.

In the end, this project is a reminder that behind every KPI is a person — and the goal of good People Analytics is not to reduce them to a number, but to understand them well enough to help them thrive.